Apex Consumer Electronics Pvt. Ltd.

Historical Financial Analysis, Forecasting & DCF Valuation

Independent workbook audit | FY2024–FY2026 actual; FY2027E–FY2029E forecast

Basis: supplied synthetic workbook only. INR crore for statements; INR for operating transactions. No external market data used. All principal base outputs independently rebuilt in Python; original formulas left unchanged.

1. Executive summary

Revenue and profits improve, liquidity strengthens, and modeled base FCFF rises. The independently replicated base equity value is 83.401216 at 12.789865% WACC. However, historical cash does not reconcile and scenario gross-margin wiring is defective. The dashboard is a validated static snapshot, not a repair or live scenario engine. See the separate one-page summary for portfolio presentation.

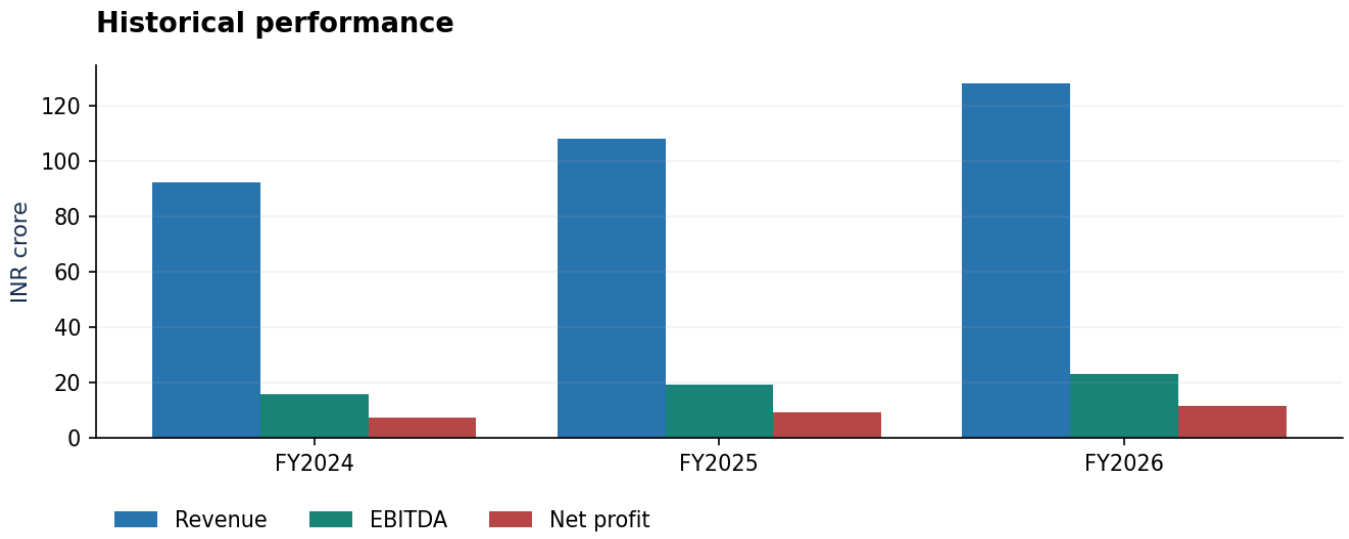
2. Company overview and data limitations

Apex is a fictional consumer-electronics business with synthetic historical statements and 360 operating transactions. The workbook offers no evidence on market share, competitive position, tax jurisdiction specifics, actual financing terms or customer concentration beyond its sample. FY2023 opening cash, working capital and capital balances are absent. Historical statements and supporting presentation tables are largely hardcoded; arithmetic consistency does not independently substantiate economic reality. Sales data are a separate sample, not a reconciled ledger. No third-party facts or valuation benchmarks are assumed.

3. Historical financial performance: FY2024–FY2026

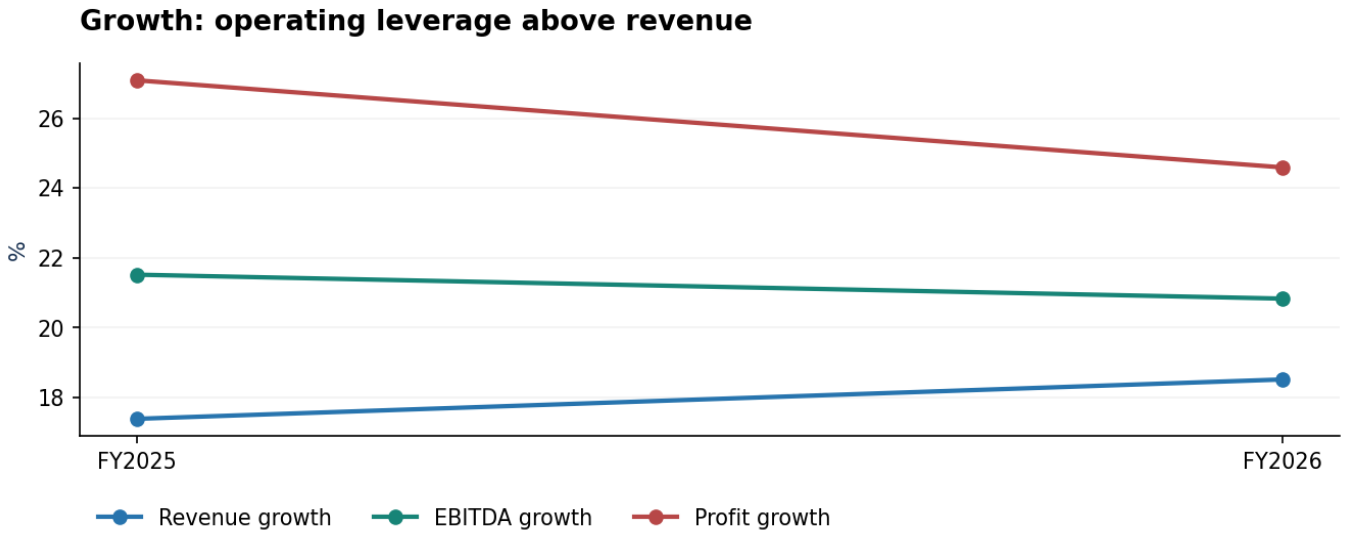
Income-statement arithmetic agrees throughout. Historical Analysis lookups and cached values agree with Historical_IS, including all margins and growth calculations. Revenue, COGS and gross profit grow together, while OpEx grows more slowly. This is consistent with operating leverage, but synthetic inputs do not establish actual cost productivity. Management should obtain price-volume and cost-center bridges before treating the margin gain as recurring.

Year	Revenue	COGS	GP	OpEx	EBITDA	D&A	EBIT	Interest	NP
FY2024	92.000	55.200	36.800	21.000	15.800	3.800	12.000	2.400	7.200
FY2025	108.000	64.800	43.200	24.000	19.200	4.200	15.000	2.800	9.150
FY2026	128.000	76.800	51.200	28.000	23.200	4.800	18.400	3.200	11.400



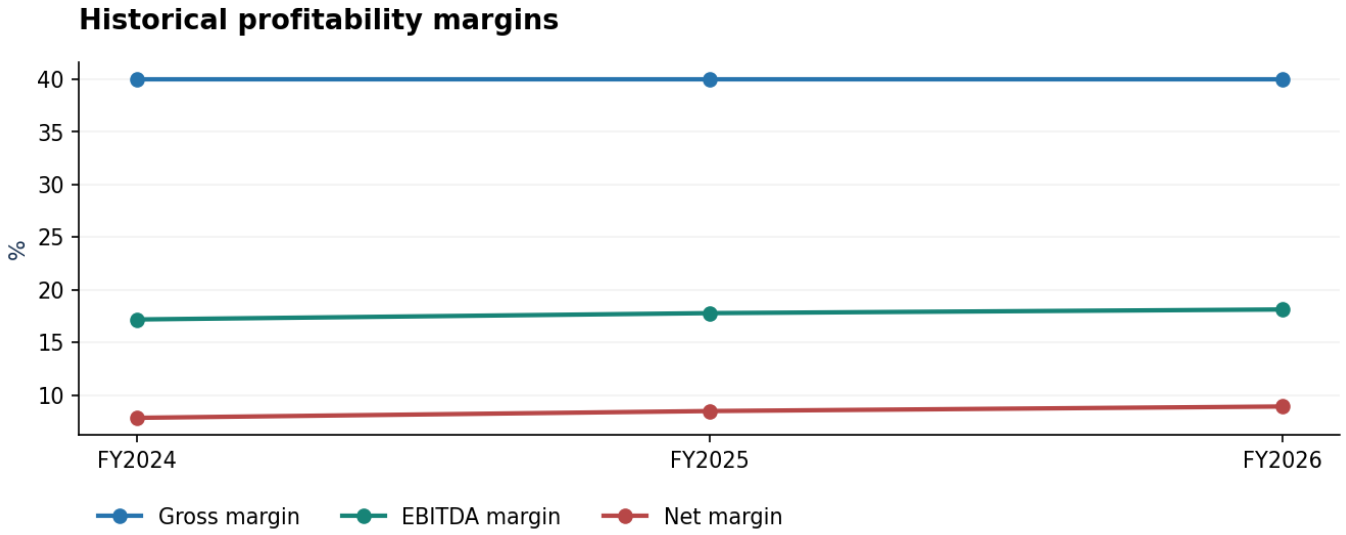
4. Revenue and growth analysis

Revenue rises 17.39% in FY2025 and 18.52% in FY2026, producing 17.95% two-year CAGR. EBITDA grows 21.52%/20.83%; net profit grows 27.08%/24.59%. Profit growth exceeding revenue supports operating leverage, rather than evidence of higher product gross margins. Separate pricing, volume and mix drivers are not available. Investigate whether sales growth can continue without stretching credit terms or inventory.



5. Gross profit and margin analysis

Gross profit increases from 36.8 to 51.2, with gross margin fixed at 40% in every historical year. The increase is entirely consistent with revenue scaling; there is no modeled gross-margin improvement. Historical product mix, procurement and pricing explanations would be hypotheses, not findings. The product master targets and transaction-level margins are different datasets and should not be substituted for the 40% statement margin.



6. Operating profitability and cost structure

OpEx/revenue falls from 22.83% to 21.88%, lifting EBITDA margin from 17.17% to 18.13%. EBIT margin rises from 13.04% to 14.38%; net margin rises from 7.83% to 8.91%. Interest rises from 2.4 to 3.2 but earnings rise faster. The model applies an effective 25% historical tax rate consistently. Investigate fixed versus variable costs: the dataset supports mathematical leverage, not a causal claim about automation, headcount or procurement.

7. Liquidity and solvency

All historical balance sheets balance; current assets are 42/49.5/59, assets 72/84/99, liabilities 37/42/47 and equity 35/42/52. Broad quick ratio includes other current assets; stricter cash-plus-AR coverage is 1.375x/1.405x/1.524x. Debt-to-total-assets is incorrectly labeled/calculated in the workbook using current assets: corrected values are 25.00%/23.81%/22.22%. ROA uses average assets in FY2025/FY2026 and closing assets in FY2024 due to missing opening data; ROE has the analogous

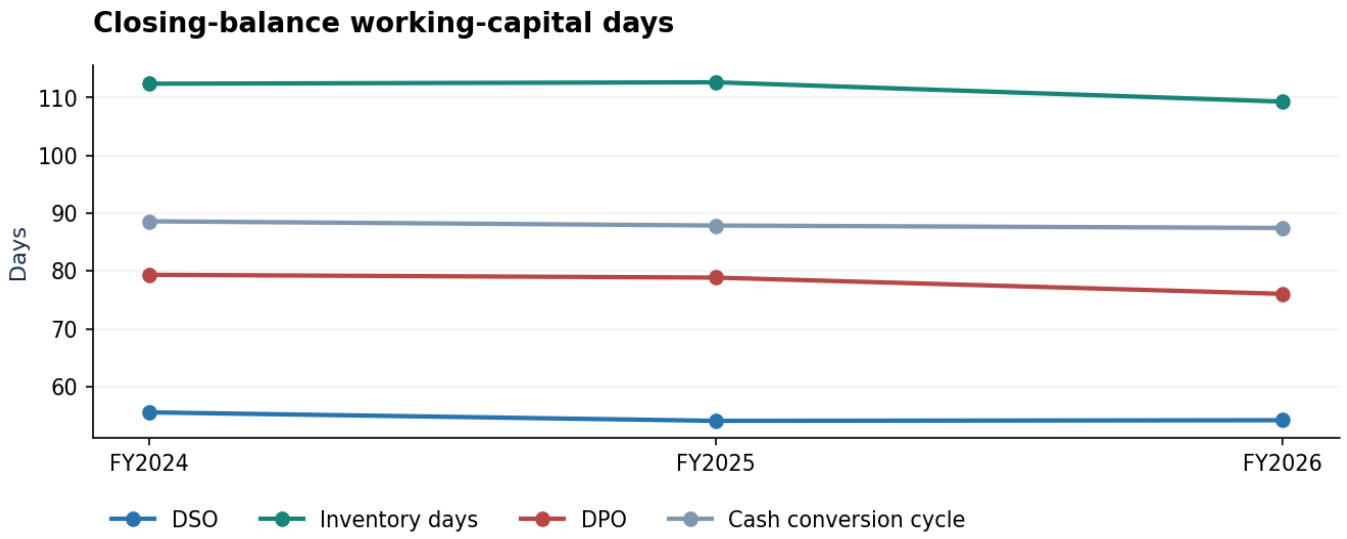
convention. Stronger ratios do not resolve unknown receivable collectability or debt maturities.

Year	Current ratio	Quick ratio	Cash ratio	Debt/equity	Debt/EBITDA	Interest coverage	ROA	ROE
FY2024	2.625	1.562	0.500	0.514	1.139	5.000	0.100	0.206
FY2025	2.676	1.595	0.541	0.476	1.042	5.357	0.117	0.238
FY2026	2.810	1.714	0.619	0.423	0.948	5.750	0.125	0.243

8. Working-capital efficiency and cash conversion cycle

Operating NWC (AR + inventory – AP) is 19/22/26, distinct from total current assets minus current liabilities of 26/31/38. Closing-balance CCC falls only 1.16 days across the historical period. DPO falls from 79.35 to 76.04 days, offsetting part of the inventory improvement. On average balances, FY2025/FY2026 CCC is 81.67/80.79 days, DSO 50.69/49.90, inventory 104.21/102.18 and DPO 73.23/71.29. Both conventions indicate modest improvement; they must not be mixed. FY2024 average-balance metrics are unverified without FY2023 data. Request aging and inventory obsolescence schedules.

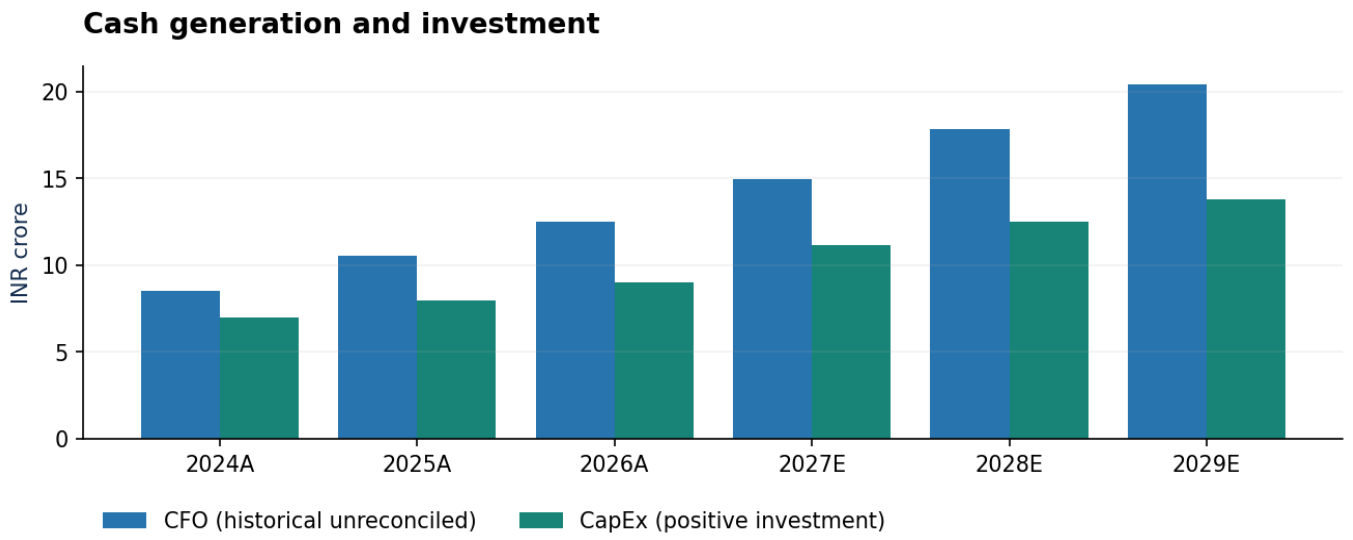
Year	DSO	Inventory days	DPO	CCC	Operating NWC
FY2024	55.543	112.409	79.348	88.605	19.000
FY2025	54.074	112.654	78.858	87.870	22.000
FY2026	54.180	109.310	76.042	87.448	26.000



9. Historical cash flow analysis

CFO arithmetic recomputes to 8.5/10.55/12.5, CFI to –7/–8/–9, and CFF to 0.5/0.5/0. CFO less CapEx is 1.5/2.55/3.5, a levered cash proxy—not FCFF. CFO/net profit declines from 1.18x to 1.10x as growth absorbs cash. Net cash movement is 2/3.05/3.5, versus observable BS changes of unavailable/2/3. FY2024 would imply opening cash 6, but this is not verified. FY2025 and FY2026 residuals are +1.05/+0.50; no missing flows were invented. Other working-capital CF adjustments +0.2/+0.3 are not explained by offsetting OCA/OCL changes. PPE roll-forwards have +0.2 annual residuals, while equity bridges have –0.65/+0.60 residuals after reported dividends. Request full reconciliations before cash-quality conclusions.

Year	CFO	CFI	CFF	CF cash change	BS cash change
FY2024	8.500	-7.000	0.500	2.000	Unverified
FY2025	10.550	-8.000	0.500	3.050	2.000
FY2026	12.500	-9.000	0.000	3.500	3.000



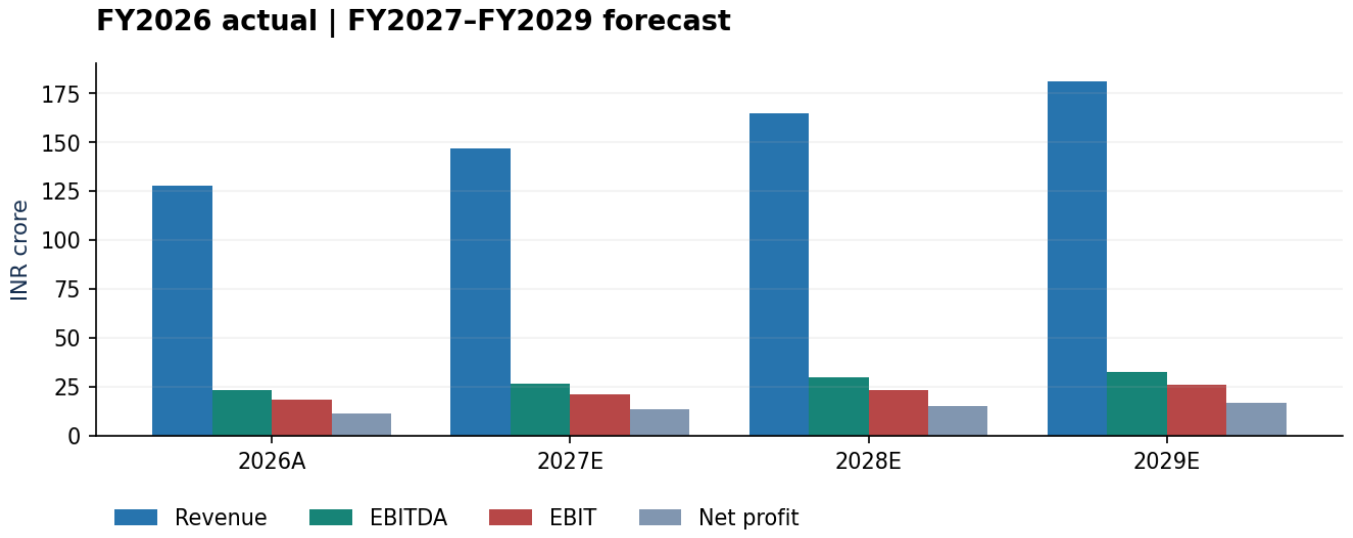
10. FY2027E–FY2029E forecast assumptions

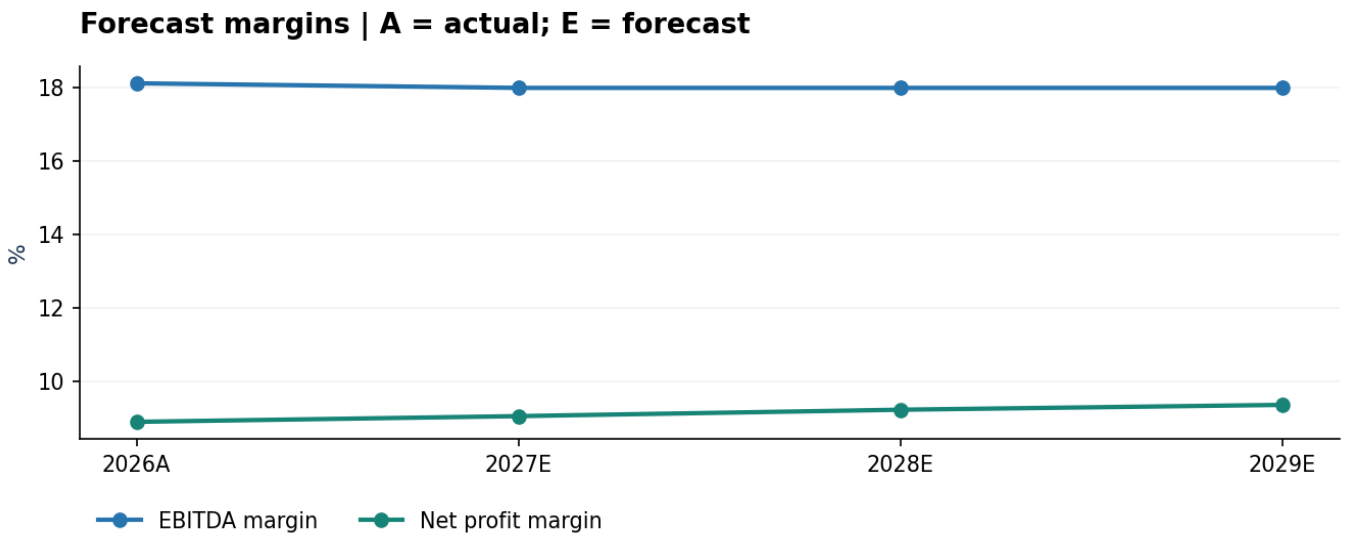
Selected scenario is Base. Growth is 15%/12%/10%; gross margin 40%; OpEx 22% of revenue; D&A 3.75%; tax 25%; DSO 55; inventory days 110; DPO 78; CapEx 7.6%; interest 14.5% on 22 debt. Forecast growth is below recent actual growth. EBITDA margin is 18%, slightly below FY2026 actual 18.125%, not an expansion story. Assumptions are inputs, not externally corroborated forecasts. Gross-margin, tax, depreciation, CapEx, interest and funding links require governance improvements. Zero dividends, zero borrowing and constant other balances are implicit simplifications.

11. Forecast income statement

The independently rebuilt base case agrees with source cached outputs. FY2029 net profit is (25.842432 – 3.19) × 75% = 16.989324. Dashboard_DATA G4 has 16.57; there is no formula supporting that extract or evidence explaining its origin. Tax is 4.4465/5.07578/5.663108. Net margin improves to 9.37% because fixed interest becomes a smaller share of revenue, while EBITDA margin stays 18%. Validate cost flexibility and funding needs before relying on this leverage.

Year	Revenue	EBITDA	EBIT	Interest	Tax	Net_Profit
FY2027E	147.200	26.496	20.976	3.190	4.446	13.339
FY2028E	164.864	29.676	23.493	3.190	5.076	15.227
FY2029E	181.350	32.643	25.842	3.190	5.663	16.989





12. Forecast balance sheet and cash flow

AR, inventory and AP independently recompute from annual revenue/COGS and day assumptions using 365 days. CFO is net income + D&A – change in operating NWC; other current balances stay fixed. PPE equals prior PPE + CapEx – D&A, and equity retains all profits. Forecast cash accumulates to 28.693409; all three forecast balance-sheet checks are zero within floating-point precision. This confirms base arithmetic under the simplifying assumptions, not that financing adequacy is proven. CFF is hardcoded zero and not derived from financing line items; introduce funding and dividend controls before general scenario use.

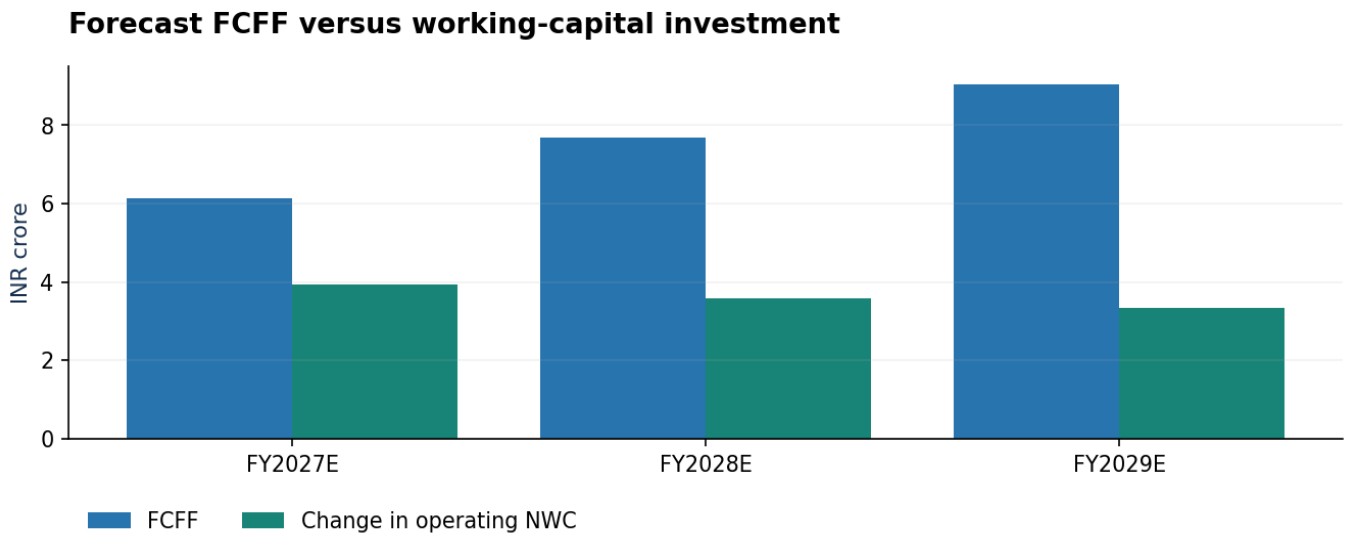
Year	AR	Inventory	AP	NWC	PPE	Cash	Debt	Equity	Assets	BS_Check
FY2027E	22.181	26.617	18.874	29.924	39.667	16.748	22.000	65.340	115.213	0.000
FY2028E	24.843	29.811	21.139	33.515	46.014	22.038	22.000	80.567	132.706	-0.000
FY2029E	27.327	32.792	23.253	36.866	52.996	28.693	22.000	97.556	151.809	0.000

Year	Beginning_Cash	CFO	CFI	CFF	Change_Cash	Cash
FY2027E	13.000	14.936	-11.187	0.000	3.748	16.748
FY2028E	16.748	17.819	-12.530	0.000	5.289	22.038
FY2029E	22.038	20.438	-13.783	0.000	6.656	28.693

13. FCFF analysis

NOPAT equals EBIT × (1 – tax); FCFF equals NOPAT + D&A – CapEx – change in operating NWC. Forecast FCFF rises 6.140855 → 7.681703 → 9.048352. CapEx exceeds D&A, but incremental NWC declines as growth slows. FCFF exceeds forecast cash accumulation by after-tax interest of 2.3925 each year, illustrating the unlevered versus levered distinction. Historical FCFF is 4.45/5.60 for FY2025/FY2026 using core operating NWC only; FY2024 is unverified because opening NWC is absent and its source FCFF lookup searches a nonexistent income-statement label. Do not plot a fabricated FY2024 value.

Year	EBIT	NOPAT	DA	CapEx	Delta_NWC	FCFF
FY2027E	20.976	15.732	5.520	11.187	3.924	6.141
FY2028E	23.493	17.620	6.182	12.530	3.591	7.682
FY2029E	25.842	19.382	6.801	13.783	3.351	9.048



14. WACC calculation and DCF valuation

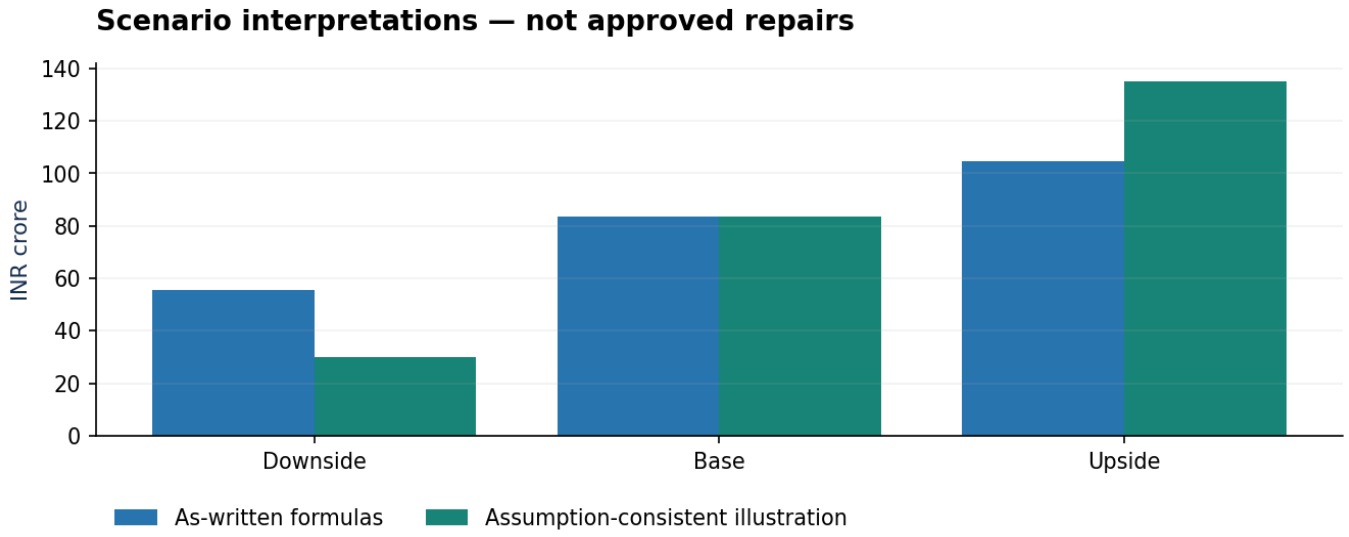
CAPM: 7% risk-free + 1.1 beta × 6% equity premium = 13.60% cost of equity. After-tax debt cost is 14.5% × 75% = 10.875%. Debt/equity weights are 22/74 and 52/74 (29.7297% / 70.2703%), producing WACC 12.7898648649%. Book equity is used; this is not an independently established market or target capital structure. Year-end discount factors are 0.886604485 / 0.786067512 / 0.696930981. PV explicit FCFF is 17.788923. With 4% growth below WACC, terminal value is 107.058367 and its PV is 74.612293. EV is 92.401216; adding FY2026 cash 13 and subtracting debt 22 gives equity 83.401216. End-FY2026 is the implied valuation date; forecast accumulated cash is not added again. Terminal value assumes sustainable growth but does not explicitly model terminal reinvestment/ROIC; validate that relationship. All cash is assumed excess and all debt captured: both bridge assumptions need review.

Valuation component	Value
PV explicit FCFF	17.789
Terminal value	107.058
PV terminal value	74.612
Enterprise value	92.401
Cash added	13.000
Debt deducted	22.000
Equity value	83.401
Terminal share of EV	0.807

15. Scenario analysis

Two independent calculations are deliberately separated. AS-WRITTEN uses historical 40% gross margin even when the selector changes forecast gross margin. ASSUMPTION-CONSISTENT ILLUSTRATION uses 38%/40%/42% for downside/base/upside, with all other stated scenario inputs. No correction was applied to the original workbook. The static upside 104.51 matches the as-written path, so it omits the intended gross-margin uplift. Static downside 30.41 matches neither path (55.67 as-written; 29.91 illustration), indicating an additional unexplained inconsistency. These are model sensitivities, not probability-weighted valuations. Obtain approval of scenario semantics and rerun a live model before presenting a definitive range.

Scenario	Static_Equity	As_written_Equity	Assumption_consistent_Equity
Downside	30.410	55.669	29.910
Base	83.410	83.401	83.401
Upside	104.510	104.509	135.125



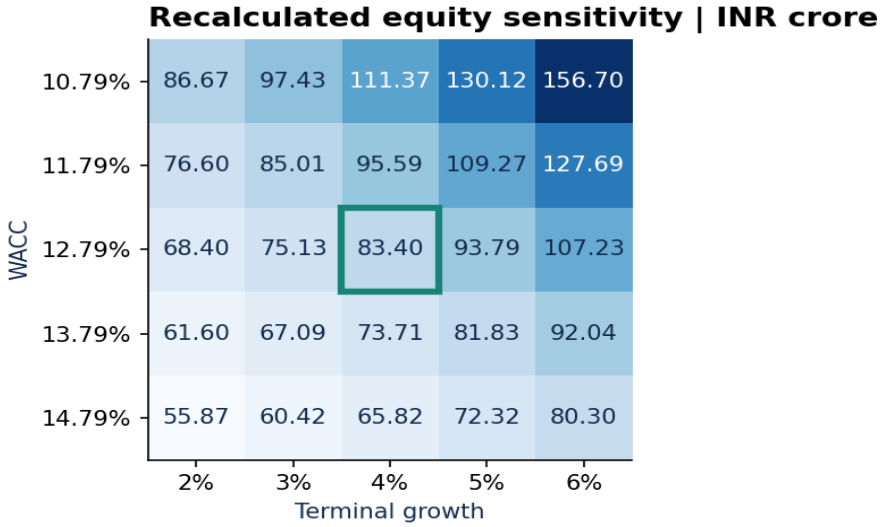
Downside illustrative FCFF (2027/28/29): -0.537184 / 3.280225 / 4.115847.

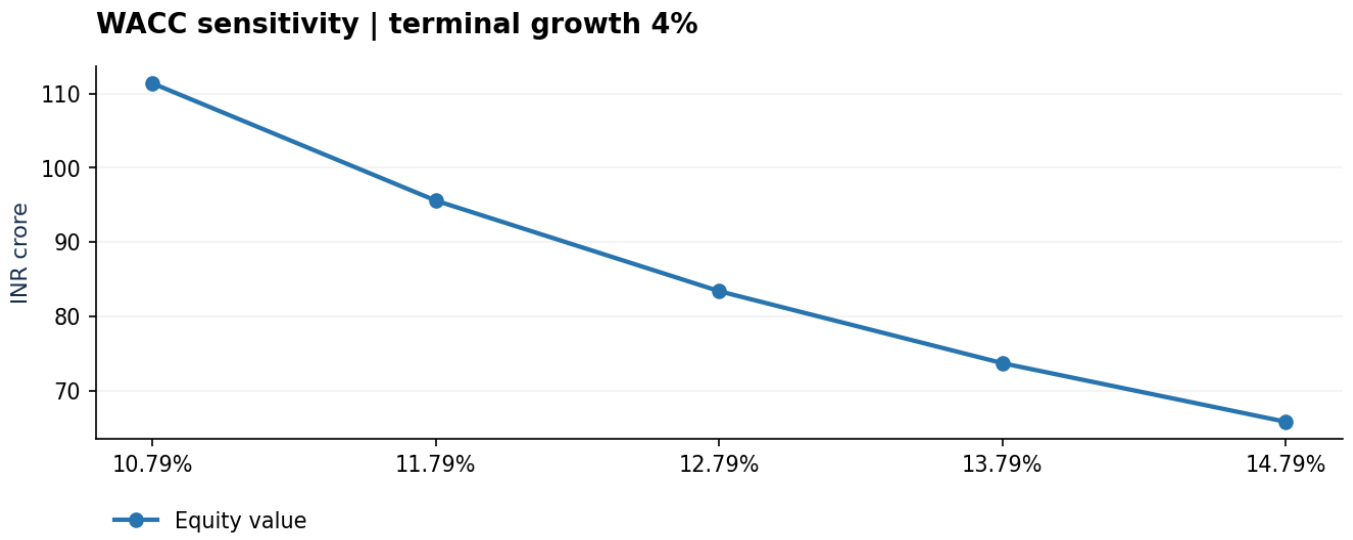
Base illustrative FCFF (2027/28/29): 6.140855 / 7.681703 / 9.048352.

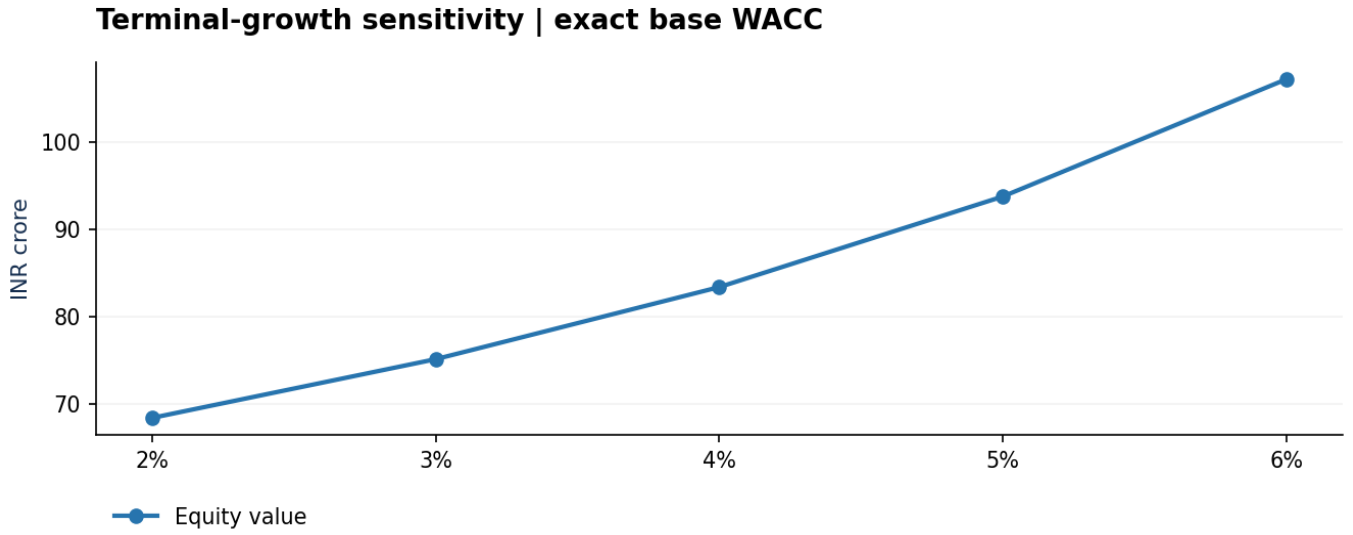
Upside illustrative FCFF (2027/28/29): 12.207513 / 11.656887 / 13.881261.

16. Sensitivity analysis and valuation risks

All 25 two-variable cells and both one-variable tables were recalculated, using the existing header grids. The two-variable WACC rows are rounded inputs (10.79%–14.79%); the highlighted 12.79% / 4% intersection is about 83.40, versus source static 83.41. Exact main WACC gives 83.401216. The rounded-rate difference is tiny; some static-cell discrepancies exceed 0.02 and require more than output rounding alone. Full cell differences are in the validation register. The independently recalculated WACC-only and growth-only charts replace static approximations for presentation. Terminal PV contributes 80.7482% of EV: a three-year explicit forecast leaves most value dependent on perpetual growth and discount rate. No sensitivity value represents a guaranteed outcome.







17. Model validation findings

408 checks and observations are recorded with source cell references, reported outputs, independent values, differences, severity and proposed action. Passing arithmetic rows are marked Low, not confirmed errors. Structural warnings with no meaningful numerical recalculation are explicitly identified. Confirmed errors: dashboard profit, debt/assets denominator, scenario gross-margin wiring, cash reconciliation and static downside mismatch. Potential improvements: average-balance turnover, stricter quick ratio, market/target capital weights, explicit debt/dividend schedules. No cached Excel error cells or literal #REF! formulas were found; that does not prove all peripheral formulas will recalculate in every Excel version. XLOOKUP/spilled arrays require compatible Excel. All existing non-Dashboard source worksheet XML, including cached results, is preserved byte-for-byte in the deliverable copy. Workbook was not recalculated by Excel; selected principal metrics were independently recalculated instead.

18. Key business risks and limitations

Growth execution and margin assumptions are not substantiated by market evidence. Working-capital funding and credit risk remain important despite positive modeled cash. Historical cash and equity/PPE residuals weaken confidence in cash-conversion narratives. Constant debt, no dividends and fixed other balances simplify the forecast. Book-weight WACC, short forecast horizon and terminal economics dominate valuation uncertainty. Synthetic transaction revenue does not reconcile to statements: it is 0.913385/0.937947/0.935378 crore versus 92/108/128 crore. Transaction COGS and gross profit are stored values, not linked to product-master costs. Quantity × price × (1 – discount) does not consistently equal reported revenue, so a new gross-profit figure should not be inferred by substituting target margins. Product-master implied standard costs are a possible benchmark, not proven actual costs. The revenue arithmetic residual totals approximately INR 1,609,843.45 across the sample. Calendar-year/quarter labels in Sales_Clean must not be mistaken for April–March fiscal periods. Supporting sales data are excluded from financial KPI construction.

FiscalYear	Revenue_Crore	Statement_Revenue_Crore
2,024.000	0.913	92.000
2,025.000	0.938	108.000
2,026.000	0.935	128.000

19. Overall analyst conclusion

The selected Base case is arithmetically reproducible and the base DCF agrees to full precision. Historical earnings and financial position show improvement, while modeled FCFF strengthens as growth slows. Nonetheless, cash reconciliation and scenario controls are not yet production-ready. The appropriate conclusion is conditional confidence in the independently replicated base arithmetic—not unconditional confidence in the whole workbook, a definitive scenario range, or an investment recommendation.

20. Recommended next steps

Priority 1: reconcile opening/closing cash and historical working capital, PPE and equity without plugs. Priority 2: approve COGS links to forecast gross margin, tax and depreciation links, CapEx/interest assumption references, and explicit debt/dividend controls. Priority 3: replace static scenario and sensitivity summaries with transparent calculation outputs after validation. Priority 4: test formula compatibility and scenario switching in current Excel. Priority 5: refresh validated dashboard snapshots and run acceptance tests against every KPI and chart. Preserve the uploaded original as an immutable audit baseline. The delivered workbook changes only Dashboard and adds Audit_Validated_Data, Audit_Validation_Register and Audit_Notes; Dashboard_DATA remains unchanged and flagged.